

A chemical genetic approach reveals distinct EphB signaling mechanisms during brain development

Michael J Soskis^{1,5}, Hsin-Yi Henry Ho^{1,5}, Brenda L Bloodgood¹, Michael A Robichaux², Athar N Malik¹, Bulent Ataman¹, Alex A Rubin¹, Janine Zieg¹, Chao Zhang^{3,4}, Kevan M Shokat³, Nikhil Sharma¹, Christopher W Cowan^{2,4} & Michael E Greenberg¹

EphB receptor tyrosine kinases control multiple steps in nervous system development. However, it remains unclear whether EphBs regulate these different developmental processes directly or indirectly. In addition, given that EphBs signal through multiple mechanisms, it has been challenging to define which signaling functions of EphBs regulate particular developmental events. To address these issues, we engineered triple knock-in mice in which the kinase activity of three neuronally expressed EphBs can be rapidly, reversibly and specifically blocked. We found that the tyrosine kinase activity of EphBs was required for axon guidance *in vivo*. In contrast, EphB-mediated synaptogenesis occurred normally when the kinase activity of EphBs was inhibited, suggesting that EphBs mediate synapse development by an EphB tyrosine kinase-independent mechanism. Taken together, our data indicate that EphBs control axon guidance and synaptogenesis by distinct mechanisms and provide a new mouse model for dissecting EphB function in development and disease.

The EphB family of receptor tyrosine kinases (RTKs) are critical regulators of cell-cell contacts in the developing nervous system, mediating processes as diverse as axon guidance, topographic mapping, neuronal migration and synapse formation^{1–3}. In addition to these developmental roles, EphB dysfunction in the mature organism contributes to pathologies such as cancer, Alzheimer's disease and, possibly, autism^{4–8}. The signaling mechanisms underlying EphB-mediated development and disease are largely unknown.

As the EphB family of receptors has been shown to regulate a large number of developmental processes, it has been particularly difficult to determine the specific functions of EphBs at defined times during brain development. The presence of at least three partially redundant EphB family members in the nervous system further complicates investigation into the biological functions of EphB proteins. For example, *Ephb1*, *Ephb2* and *Ephb3* single and compound mutant mice have defects in a number of processes, including stem-cell proliferation, axon guidance, filopodial motility, dendritic spine formation, synapse development and long-term potentiation, but it is unclear which of these interdependent phenotypes are direct and which are secondary to the disruption of EphB signaling at an earlier developmental step^{9–13}.

Another major hurdle in understanding the function of EphBs is the complex nature of their signaling capabilities. EphBs can engage in bidirectional signaling with their transmembrane ligands, the ephrin-Bs. In the forward direction of signaling, the interaction of clustered ephrin-B ligands on one cell with EphB receptors on another leads

to EphB oligomerization and auto-phosphorylation, the induction of EphB kinase activity, and the recruitment of cytoplasmic proteins via SH2-binding and PDZ-binding motifs of EphBs¹⁴. In addition, the extracellular region of EphBs, which contains fibronectin repeat domains, can recruit binding partners such as subunits of the NMDA subtype of glutamate receptor^{15,16}. In the reverse direction of EphB and ephrin-B signaling, phosphorylation of the cytoplasmic tail of ephrin-Bs results in the recruitment of SH2 domain-containing proteins and initiation of downstream signal transduction¹⁷. Thus, through a complex array of potential signaling pathways, EphBs are able to mediate a wide range of processes during nervous system development.

For the most part, it remains unknown which cellular processes require EphB RTK activity and which cellular responses are mediated by EphB tyrosine kinase-independent signaling events. Cytoplasmic deletions of EphBs have been used to assess the requirement of the intracellular domain in mediating specific EphB-regulated processes, but this approach fails to distinguish kinase activity from other modes of cytoplasmic signaling¹⁸. In particular, given that ephrin-B binding to EphBs induces the formation of EphB oligomers in the plasma membrane, it remains a likely possibility that EphB oligomerization and scaffolding, in the absence of induction of EphB tyrosine kinase activity, mediates some of the biological effects of EphBs¹⁴. Thus, new ways of selectively inhibiting specific functions of EphBs are critically needed to clarify the kinase-dependent and kinase-independent mechanisms by which EphBs control specific developmental events, such as axon guidance and synapse formation.

¹Department of Neurobiology, Harvard Medical School, Boston, Massachusetts, USA. ²Department of Psychiatry, University of Texas Southwestern Medical Center, Dallas, Texas, USA. ³Department of Cellular and Molecular Pharmacology, University of California, San Francisco, California, USA. ⁴Present address: Department of Chemistry, University of Southern California, Los Angeles, California, USA (C.Z.), Department of Psychiatry, McLean Hospital, Harvard Medical School, Belmont, Massachusetts, USA (C.W.C.). ⁵These authors contributed equally to this work. Correspondence should be addressed to M.E.G. (michael_greenberg@hms.harvard.edu).

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Much of what is currently known about the role of EphB signaling during axon guidance *in vivo* comes from studies of retinal and cortical axon tracts. Notably, genetic deletion of individual or combinations of EphB family members cause marked axon guidance defects that result in the abnormal formation of several axon tracts, including the ipsilateral retinocollicular projection and axonal tracts in the corpus callosum and the anterior commissure^{9,19}. However, it remained to be determined whether EphB-dependent axon guidance decisions are mediated by the kinase activity of EphBs or by other modes of EphB signaling, such as PDZ-domain interactions, cytoplasmic domain oligomerization, reverse signaling through ephrin-Bs or EphB extracellular domain interactions.

In general, the mechanisms by which the cytoplasmic domains of axon guidance receptors signal growth cone attraction or repulsion have been difficult to identify. Most axon guidance receptors that have been studied to date, such as Robo, DCC, plexins and neuropilins, do not possess intrinsic kinase activity, suggesting that the predominant mode of axon guidance signaling may be kinase independent²⁰. With regard to EphBs, studies in the visual system have suggested a role for the kinase activity, whereas studies in the cortex have suggested kinase-independent roles of EphBs^{9,21–24}. However, none of these experiments were able to directly address the requirement of the kinase activity of EphBs when they are expressed at endogenous levels *in vivo*.

In addition to their function in axon guidance, EphBs have been shown to be important for cortical and hippocampal synapse formation. *Ephb* knockout mice or knock-in mice with cytoplasmic domain deletions display defects in dendritic spine and synapse development in dissociated neuronal cultures and in hippocampal slices^{13,18}. Overexpression of kinase-defective dominant-negative mutants also results in abnormal spine and synapse development, suggesting that the tyrosine kinase activity of EphBs is involved in these processes^{13,15,25}. However, these studies did not directly test the requirement of the kinase activity of EphBs in the regulation of synapse formation under conditions in which ephrin-Bs and EphBs are expressed at physiological levels, thereby complicating the interpretation of the findings and leaving open the possibility that aspects of EphB-dependent synapse development may be kinase independent.

To address the importance of the kinase activity of EphBs for axon guidance and synapse development, we combined chemical biology with mouse genetic engineering to reversibly inhibit EphB tyrosine kinase signaling in cultured neurons, brain slices and live animals. By mutating a bulky gatekeeper residue in the ATP-binding pocket of EphBs to a smaller alanine or glycine, we rendered the enzymatic activity of these kinases sensitive to reversible inhibition by derivatives of the Src inhibitor PP1. These analogs of PP1 do not fit into the ATP-binding pocket of wild-type kinases^{26,27} and therefore do not inhibit wild-type EphBs or other kinases. In a previous study, this approach was successfully employed to regulate Trk RTK signaling²⁸. We reasoned that this approach would be particularly well suited for investigating the more complex EphB family, allowing us to inhibit the kinase activity of EphBs while preserving extracellular interactions, reverse signaling or other cytoplasmic domain functions of EphBs.

We generated a triple knock-in mouse line that harbors ATP-binding pocket mutations in EphB1, EphB2 and EphB3. By specifically blocking the kinase function of EphBs in these knock-in mice in a temporally controlled manner, we found for the first time, to the best of our knowledge, a requirement for the kinase activity of EphBs in axonal guidance *in vivo*. In contrast, the kinase activity of EphBs was not required for another key EphB-mediated event, the regulation of

excitatory synapse formation. Thus, when expressed under physiologically relevant conditions, EphBs differentially regulate key aspects of neuronal development by kinase-dependent and kinase-independent mechanisms.

RESULTS

A chemical genetic approach to studying EphB signaling

To selectively inhibit the tyrosine kinase activity of EphBs, we employed a strategy that combines the advantages of both pharmacology and genetics and in which drug sensitivity can be engineered into a protein^{29,30}. The ATP-binding pocket of all kinases contains a bulky hydrophobic gatekeeper residue that is not essential for the catalytic function of the kinase domain. However, this gatekeeper residue, when mutated to an alanine or a glycine, can render the kinase sensitive to inhibition by PP1 analogs that cannot effectively enter the wild-type ATP-binding pocket (Fig. 1a)^{29,30}. We engineered EphBs with modified gatekeeper residues and refer to these PP1 analog-sensitive EphBs as AS-EphBs.

To design AS-EphB mutants, we compared the amino acid sequence of kinase domains of EphBs with those of related tyrosine kinases for which analog-sensitive versions had been successfully made^{26,28}. This analysis revealed a gatekeeper threonine residue in the ATP-binding pocket of mouse EphB1, EphB2 and EphB3 (Fig. 1b). We substituted these residues with either alanine or glycine to generate *Ephb1*^{T697G}, *Ephb2*^{T699A} and *Ephb3*^{T706A}.

To verify that the mutations introduced into the EphB ATP-binding pockets did not affect the kinase activity of EphBs in the absence of the PP1 analogs, we assessed the activity of these kinases using a heterologous cell culture system. Activation of EphBs results in receptor auto-phosphorylation on several juxtamembrane tyrosine residues. We previously generated an antibody that specifically recognizes the phosphorylated form of these juxtamembrane tyrosine residues for all EphBs, and used the juxtamembrane tyrosine phosphorylation detected by this antibody as a readout for receptor kinase activation¹⁵. We overexpressed EphBs in HEK 293 cells in which EphBs cluster spontaneously and become activated in a ligand-independent manner. We probed lysates from cells overexpressing AS-EphBs or wild-type EphBs with antibody to phosphorylated EphB and found that wild-type EphBs and AS-EphBs were phosphorylated at their juxtamembrane tyrosines to a similar extent (Fig. 1c). This result indicates that the analog-sensitive mutation does not affect the ability of EphBs to activate their kinase domains in the absence of PP1 analogs.

To test the ability of PP1 analogs to inhibit AS-EphBs, we treated HEK 293 cells overexpressing AS-EphBs or wild-type EphBs with either of two bulky PP1 analogs, 4-amino-1-*tert*-butyl-3-(1'-naphthyl)pyrazolo[3,4-d]pyrimidine (1-NA-PP1) or 1-(*tert*-butyl)-3-(3-methylbenzyl)-1H-pyrazolo[3,4-d]pyrimidin-4-amine (3-MB-PP1) (Fig. 1a) and assessed EphB tyrosine phosphorylation. We found that incubation with 1-NA-PP1 (250 nM) or 3-MB-PP1 (1 μ M) blocked the phosphorylation of AS-EphBs, but not the phosphorylation of wild-type EphBs (Fig. 1c). The drug vehicle DMSO alone did not have any effect on either wild-type EphBs or AS-EphBs. These results indicate that the kinase activity of AS-EphBs was selectively inhibited by PP1 analogs. Inhibition of kinase activity of AS-EphBs was rapid, occurring in minutes (Fig. 1d). Given that PP1 analogs act competitively, inhibition was readily reversible following removal of 1-NA-PP1 (Fig. 1e). The specificity, rapidity and reversibility of the kinase inhibition make this chemical genetic strategy an ideal platform for studying dynamic biological processes in cells and animals.

To quantify the potency and specificity of 1-NA-PP1 and 3-MB-PP1 with respect to inhibition of EphB1, EphB2 and EphB3, we conducted

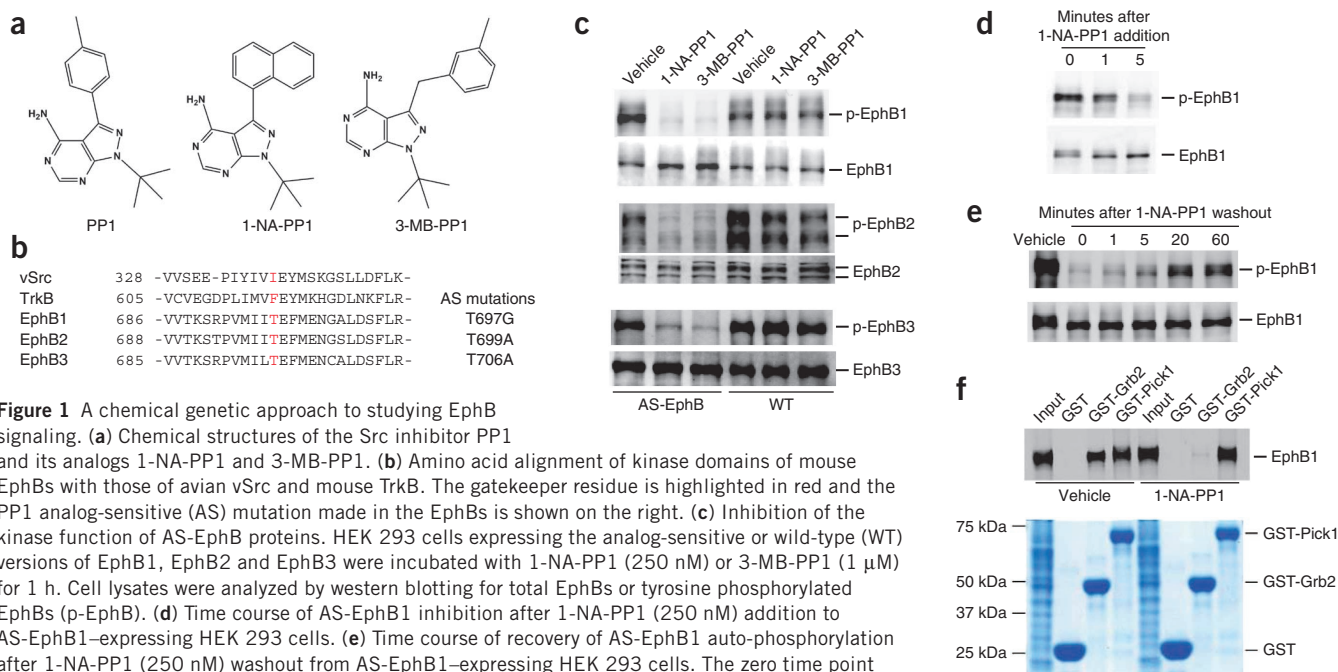


Figure 1 A chemical genetic approach to studying EphB signaling. **(a)** Chemical structures of the Src inhibitor PP1 and its analogs 1-NA-PP1 and 3-MB-PP1. **(b)** Amino acid alignment of kinase domains of mouse EphBs with those of avian vSrc and mouse TrkB. The gatekeeper residue is highlighted in red and the PP1 analog-sensitive (AS) mutation made in the EphBs is shown on the right. **(c)** Inhibition of the kinase function of AS-EphB proteins. HEK 293 cells expressing the analog-sensitive or wild-type (WT) versions of EphB1, EphB2 and EphB3 were incubated with 1-NA-PP1 (250 nM) or 3-MB-PP1 (1 μ M) for 1 h. Cell lysates were analyzed by western blotting for total EphBs or tyrosine phosphorylated EphBs (p-EphB). **(d)** Time course of AS-EphB1 inhibition after 1-NA-PP1 (250 nM) addition to AS-EphB1-expressing HEK 293 cells. **(e)** Time course of recovery of AS-EphB1 auto-phosphorylation after 1-NA-PP1 (250 nM) washout from AS-EphB1-expressing HEK 293 cells. The zero time point reflects the moment of 1-NA-PP1 washout after an initial 1-h incubation. **(f)** Effect of 1-NA-PP1 on the ability of AS-EphB1 to bind Grb2 or Pick1. HEK 293 cells expressing AS-EphB1 were incubated with either vehicle or 1-NA-PP1. Cell lysates were incubated with GST, GST-Grb2 or GST-Pick1 proteins immobilized on glutathione beads. Proteins bound to the beads were analyzed by western blotting for EphB1 (top of gel). The same gel (bottom) was stained with Coomassie blue to verify that similar amounts of GST fusion proteins were present in the binding reactions. Uncropped blots are shown in **Supplementary Figure 7**.

dose-response analyses. We determined the half maximal inhibitory concentrations of kinase inhibition, which revealed a preference of the PP1 analogs for inhibiting analog-sensitive kinases (9–48 nM) over wild-type kinases (0.6–8.9 μ M) by two orders of magnitude (**Supplementary Fig. 1a,b**). These results indicate that the kinase activity of AS-EphBs was selectively inhibited by PP1 analogs.

To confirm that binding of PP1 analogs to AS-EphBs does not affect kinase-independent aspects of EphB function, we performed *in vitro* binding assays to compare the effect of 1-NA-PP1 on tyrosine kinase-dependent and tyrosine kinase-independent protein-protein interactions. The SH2 and SH3 domain-containing adaptor protein Grb2 is a classic example of a signaling molecule that interacts with RTKs, including EphBs, through a tyrosine kinase-dependent mechanism. Following receptor activation and auto-phosphorylation, the phospho-tyrosine residues in the cytoplasmic tails of RTKs, including EphBs, recruit Grb2 through its SH2 domain^{31,32}. In contrast, binding of the PDZ domain-containing protein Pick1 to EphBs occurs through the C-terminal PDZ domain-binding motif of EphBs and does not require the tyrosine kinase activity of EphBs³³. In the binding experiments, we found that EphB1 bound strongly to both GST-Grb2 and GST-Pick1, but not to the negative control, GST alone (**Fig. 1f**). Notably, although 1-NA-PP1 (1 μ M) completely abolished the kinase-dependent interaction between AS-EphB1 and GST-Grb2, the inhibitor treatment had no effect on the kinase-independent interaction between AS-EphB1 and GST-Pick1 (**Fig. 1f**). Thus, these observations provide evidence that 1-NA-PP1 specifically targets kinase-dependent functions of EphB proteins.

Generation and validation of AS-EphB TKI mice

Encouraged by these initial experiments, we generated knock-in mice harboring the gatekeeper mutations in EphB1 (T697G), EphB2 (T699A) and EphB3 (T706A), the three catalytically active EphB

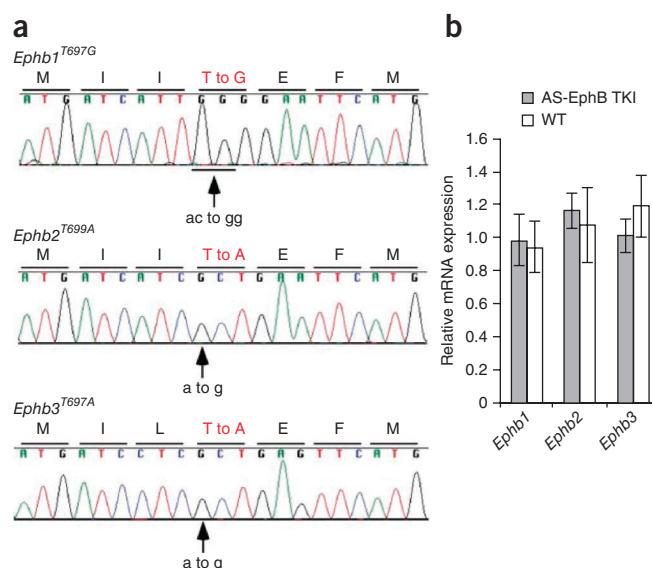
RTKs expressed in the developing brain. *Ephb1*^{T697G}, *Ephb2*^{T699A} and *Ephb3*^{T706A} single mutant mice were generated individually by homologous recombination in mouse embryonic stem (ES) cells (**Supplementary Fig. 2a–c**). ES cell clones were confirmed by sequencing the targeted alleles in ES cells (**Fig. 2a**).

Previous studies have found substantial functional redundancy of EphBs in several contexts^{18,22,34}. To overcome potential compensation by different EphB family members, we intercrossed *Ephb1*^{T697G}, *Ephb2*^{T699A} and *Ephb3*^{T706A} single mutants to generate a line that is triply homozygous for each of the targeted EphB alleles, which we refer to as EphB triple knock-in (AS-EphB TKI) mice.

Although *Ephb1*, *Ephb2* and *Ephb3* triple knockout mice suffer from marked developmental defects, including morphological abnormalities of the palate and the anogenital region^{35,36}, AS-EphB TKI mice developed normally into healthy, fertile adults, indistinguishable from wild-type mice. Notably, AS-EphB TKI brains exhibited normal morphology and were of equal size to wild-type mouse brains. *In vitro* assays measuring axon guidance, neuronal morphology and synaptic development revealed no differences between wild-type and AS-EphB TKI mice (as shown below), suggesting that AS-EphBs function normally in the absence of PP1 analogs.

Critical to the interpretation of experiments comparing wild-type and AS-EphB TKI mice is evidence that EphB mRNA expression, trafficking and ligand-mediated receptor activation occur normally in AS-EphB TKI neurons in the absence of PP1 analogs. First, we performed quantitative PCR (qPCR) to measure expression of the *Ephb1*, *Ephb2* and *Ephb3* mRNAs in neurons from wild-type and AS-EphB TKI mice. We found that mRNAs isolated from the mutant mice were expressed at the same levels as in wild-type mice (**Fig. 2b**). We conclude that the gene targeting did not affect the expression of *Ephb1*, *Ephb2* and *Ephb3* mRNA in AS-EphB TKI neurons.

Figure 2 Generation of AS-EphB TKI mice. (a) Sequencing reads from AS-EphB knock-in mouse ES cells showing the gatekeeper (analog sensitive) mutation in the *Ephb1*, *Ephb2* and *Ephb3* genes. The mutated amino acids are shown in red. The arrows indicate the DNA base substitutions. (b) Relative mRNA expression of *Ephb1*, *Ephb2* and *Ephb3* normalized to *Actb* in AS-EphB TKI and wild-type cultured cortical neurons. Data are mean of two biological replicates $\pm$ s.e.m.



We next asked whether neurons from AS-EphB TKI mice could engage in ephrin-B-induced signaling. We cultured dissociated cortical neurons from embryonic day 16.5 to 18.5 (E16.5–18.5) AS-EphB TKI or wild-type mice and stimulated the neurons with clustered ephrin-B1 for 30 min. Western blotting of AS-EphB TKI or wild-type lysates with the antibody to phosphorylated EphB revealed that both wild-type and AS-EphB TKI neurons exhibited robust EphB activation at similar levels, indicating that AS-EphBs are fully competent to engage in ephrin-B-induced kinase signaling in the absence of PP1 analogs (Fig. 3a).

To test whether the kinase function of endogenously expressed EphBs from AS-EphB TKI mice can be effectively and selectively inhibited by PP1 analogs, we pre-incubated cultured E16.5–18.5 neurons with vehicle or PP1 analogs for 1 h before ephrin-B1 stimulation. Treatment with 250 nM 1-NA-PP1 or 1 μ M 3-MB-PP1 completely abolished ephrin-B1-induced EphB activation in the AS-EphB TKI, but not wild-type, cells, thereby demonstrating the efficacy and selectivity of these PP1 analogs for AS-EphBs expressed at endogenous levels in neurons (Fig. 3a).

To more rigorously test the selectivity of PP1 analogs, we assessed the effect of these inhibitors on the kinase activity of EphA4, one of the closest relatives of EphBs. We stimulated wild-type cortical neurons that had been pre-incubated with 250 nM 1-NA-PP1 or 1 μ M 3-MB-PP1 with the EphA4 ligand ephrin-A1, immunoprecipitated EphA4 with an antibody to EphA4, and analyzed the immunoprecipitates for phospho-EphA4 or total EphA4. We found no inhibition of EphA4 autophosphorylation at concentrations of PP1 analogs that fully blocked AS-EphB tyrosine kinase function (Fig. 3b). The specificity of analog-sensitive inhibition between subfamilies of Eph receptors contrasts with previous unsuccessful attempts at designing selective inhibitors for EphBs relative to EphAs^{37,38}.

To determine whether PP1 analogs affect the cell surface expression or internalization of AS-EphB proteins, we performed surface biotin labeling for the EphB2 receptor after chronic inhibition from 6–12 days *in vitro* (DIV) in cortical neurons. We found equivalent levels of labeled EphB2 in vehicle and 1-NA-PP1 (1 μ M)-treated cells (Supplementary Fig. 3). This result indicates that surface expression of EphBs is not altered by the inhibitor treatment.

Kinase cascades can amplify signals greatly. To test whether inhibition of the kinase activity of EphBs effectively blocks the phosphorylation of downstream tyrosine kinase substrates, we examined phosphorylation of the well-characterized EphA and EphB substrate Vav2, a Rho family guanine nucleotide exchange factor that mediates growth cone collapse³⁹. Wild-type or AS-EphB TKI neurons were incubated in 1-NA-PP1 (250 nM) or 3-MB-PP1 (1 μ M) and stimulated with ephrin-B1 for 30 min to induce Vav2 tyrosine phosphorylation. Immunoprecipitation of Vav2, followed by western blotting with a pan-phospho-tyrosine antibody, revealed a substantial increase in tyrosine phosphorylation of Vav2 after ephrin-B1 stimulation (Fig. 3c,d). Treatment with 1-NA-PP1 (250 nM) or 3-MB-PP1 (1 μ M) selectively blocked the increase in Vav2 phosphorylation in AS-EphB TKI cells, but had no effect in wild-type cells (Fig. 3c,d). We conclude that PP1 analogs can selectively

block the kinase signaling of EphBs and the phosphorylation of their substrates in AS-EphB TKI cells.

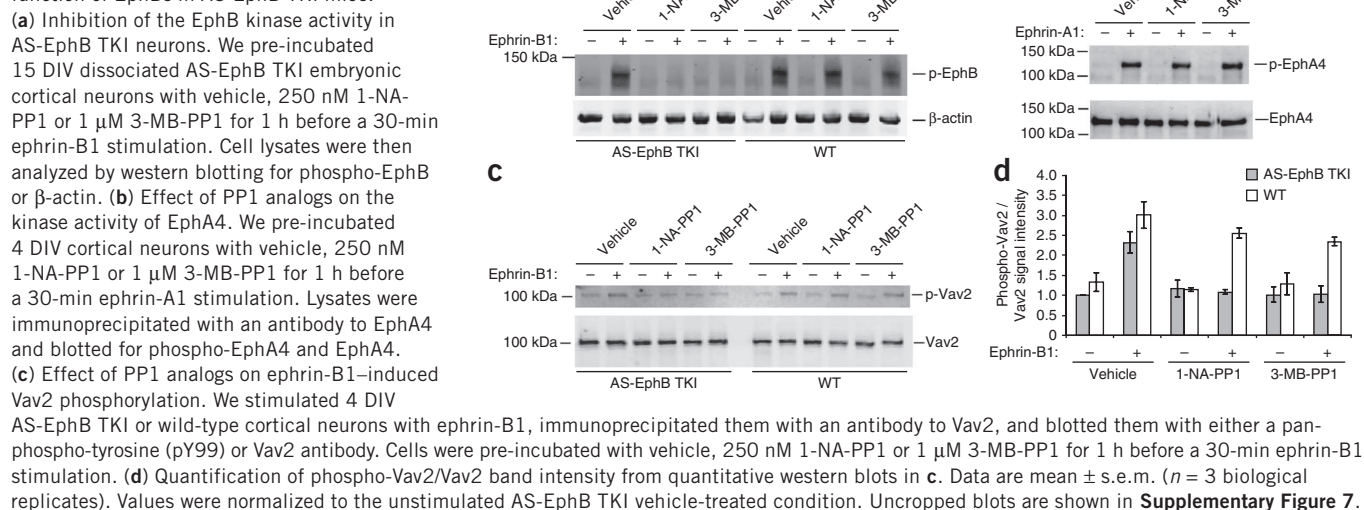
EphB RTK signaling is required for growth cone collapse

EphBs have classically been studied in the context of axon guidance^{22,40,41}. Although some studies have suggested a role for the kinase activity of EphBs in axon guidance, others have suggested kinase-independent modes of EphB signaling^{9,21–24}. To assess the role of EphB kinase activity in axon guidance, we initially chose the visual system because the importance of EphBs in this system is well established. During development of the visual system, most retinal ganglion cell (RGC) axons emanating from the retina enter and cross the optic chiasm at the midline to form the contralateral retinal projections in the lateral geniculate nucleus and superior colliculus^{24,40}. However, RGC axons from the ventrotemporal region of the retina, which express EphB1, become repelled by the ephrin-B2-expressing glia of the optic chiasm and form the ipsilateral projection that is required for stereovision⁴². Ephrin-B2 is thought to repel EphB1-expressing axons by inducing the collapse of their growth cones, an activity that can be recapitulated *in vitro*.

To begin to address whether the kinase activity of EphBs is required for repulsive axon guidance, we asked whether the kinase activity of EphBs is required for growth cone collapse. We prepared explants from E14 ventrotemporal retinae and visualized growth cones with fluorescently conjugated phalloidin (to label F-actin) and axons with antibody to neurofilament. In these explants, RGCs extended axons with broad, fan-shaped growth cones over a laminin substrate. We measured the state of growth cones by two methods. First, growth cones were scored as collapsed if they exhibited rod-like morphology and lacked visible lamellipodia. This method allowed us to quantify the percentage of growth cones that were collapsed in each explant. Second, we measured the maximum axial width of each growth cone and calculated the average axial width of the growth cones in each explant.

When treated with clustered ephrin-B2, growth cones from AS-EphB TKI explants exhibited a robust collapse response (Fig. 4a,b). Treatment of AS-EphB TKI explants with 1-NA-PP1 (250 nM) or 3-MB-PP1 (1 μ M) led to a marked decrease in the percentage of collapsed growth cones following ephrin-B treatment (Fig. 4a,b). In addition, AS-EphB TKI explants showed an ephrin-B2-induced decrease in growth cone width in vehicle-treated explants, but only

Figure 3 Selective inhibition of the kinase function of EphBs in AS-EphB TKI mice.



a slight reduction in explants exposed to 1-NA-PP1 or 3-MB-PP1 (**Fig. 4c**). This inhibitor-dependent blockade of growth cone collapse was not seen in wild-type explants; wild-type cells showed a robust collapse regardless of the presence of inhibitors, as well as a substantial reduction in average growth cone width after stimulation (**Fig. 4a–c**). On the basis of these results, we conclude that the kinase activity of EphBs is required for ephrin-B-induced RGC growth cone collapse.

To verify that treatment of explants with 1-NA-PP1 and 3-MB-PP1 led to inhibition of ephrin-B-dependent EphB tyrosine phosphorylation in RGC axons, we stained explants with antibody to phosphorylated EphB. In the absence of stimulation, minimal phospho-EphB staining was observed in growth cones from wild-type or AS-EphB TKI explants (**Fig. 4a**). Following ephrin-B2 stimulation, punctate patterns of phosphorylated EphB staining were seen in the growth cone and along the axon (**Fig. 4a**). This staining was blocked in AS-EphB TKI explants treated with 1-NA-PP1 (250 nM) or 3-MB-PP1 (1 μ M), but not in wild-type explants, indicating that PP1 analogs selectively inhibit the kinase activity of EphBs in AS-EphB TKI RGC explants (**Fig. 4a**).

A crucial feature of this chemical genetic approach is the reversibility of kinase

inhibition. To determine whether the inhibitory effect of 1-NA-PP1 on growth cone collapse is indeed reversible, we removed 1-NA-PP1 midway through a 30-min ephrin-B2 stimulation in AS-EphB TKI explants. In the presence of 1-NA-PP1, ephrin-B-stimulated growth cones did not collapse, but did collapse rapidly after washout (**Fig. 4d**). This finding demonstrates the reversibility of PP1 analog inhibition of growth cone collapse in retinal explants from AS-EphB TKI mice and rules out a general health issue as a cause of reduced growth cone collapse.

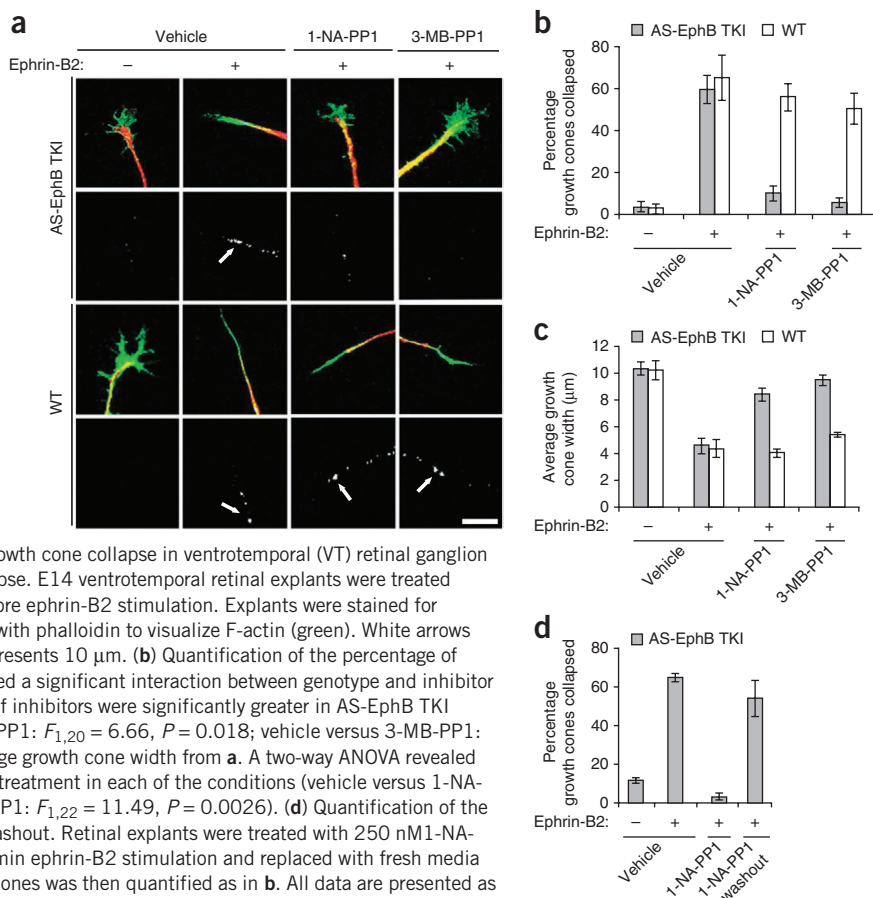
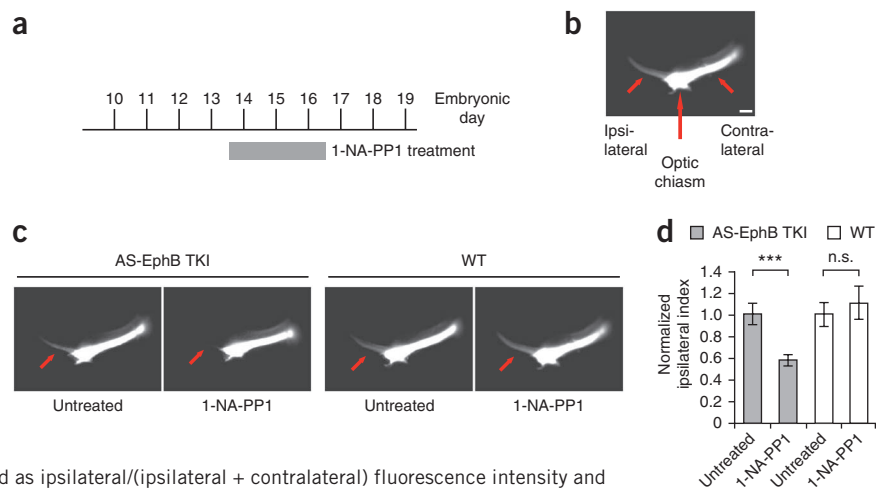


Figure 5 The kinase function of EphBs is required for the formation of the ipsilateral retinal projection *in vivo*. (a) Schedule of *in vivo* 1-NA-PP1 administration. Twice-daily subcutaneous injections of 80 mg per kg of body weight 1-NA-PP1 were administered to pregnant females from E13.5–16.5.

(b) Representative image demonstrating the orientation of the ipsilateral and contralateral retinal projections (short red arrows) of the optic tract with respect to the optic chiasm (long red arrow) as visualized by Dil labeling (white). Scale bar represents 100 μ m.

(c) Representative images of Dil-filled retinal projections at E16.5. Pregnant AS-EphB and C57BL/6 wild-type mice were treated as described in a. Red arrows denote the ipsilateral projection.

(d) Quantification of the ipsilateral phenotype shown in c. The ipsilateral index is defined as ipsilateral/(ipsilateral + contralateral) fluorescence intensity and normalized for each genotype (untreated AS-EphB TKI ($n = 30$ embryos), 1-NA-PP1-treated AS-EphB TKI ($n = 30$, embryos), untreated wild-type ($n = 15$, embryos), 1-NA-PP1-treated wild-type ($n = 17$, embryos)). *** $P < 0.001$; n.s., not significant by Student's t test. A two-way ANOVA revealed a significant interaction between genotype and inhibitor treatment, indicating that 1-NA-PP1 treatment effects were significantly greater in AS-EphB TKI embryos than in wild-type embryos ($F_{1,88} = 6.5$, $P = 0.0125$). Data are presented as mean $\pm$ s.e.m. Samples are biological replicates.



EphB RTK signaling mediates retinal axon guidance *in vivo*

Having established the requirement of the tyrosine kinase activity of EphBs for growth cone collapse *in vitro*, we next asked whether this requirement is relevant for repulsive guidance *in vivo*, where growth cones exhibit saltatory motion rather than simple extension and collapse. In addition, growth cone collapse *in vivo* is mediated by a number of cues, including membrane-bound ephrin-B rather than ectopically added aggregated soluble ephrin-Bs.

To determine whether the kinase activity of EphBs is required for the repulsive guidance of axons under physiological conditions of ephrin-B and EphB signaling *in vivo*, we examined the effect of inhibiting the kinase activity of EphBs on axon repulsion at the optic chiasm. We treated pregnant mice with 1-NA-PP1 from E13.5 to E16.5, the time at which RGC axons encounter the optic chiasm (Fig. 5a), and analyzed the retinal projections at E16.5 by Dil labeling.

To assess the degree of ipsilateral versus contralateral retinal projection, we measured the fluorescence intensity of a rectangular region in the ipsilateral projection and divided this by the total fluorescence intensity in the combined ipsilateral and contralateral projections to derive the ipsilateral index (Fig. 5b)⁴⁰. In 1-NA-PP1-treated AS-EphB TKI embryos, we found that the ipsilateral retinal projection was strongly reduced (by 42%) compared with untreated AS-EphB TKI embryos (Fig. 5c,d). In many of the AS-EphB TKI embryos treated with 1-NA-PP1, the ipsilateral projection was absent. In contrast, 1-NA-PP1 treatment had no effect on the ipsilateral retinal projection in wild-type embryos, indicating that the observed guidance deficit was a result of specific inhibition of EphBs (Fig. 5c,d). These findings suggest that the tyrosine kinase activity of EphBs is required for axon repulsion at the optic chiasm. They also illustrate the utility of AS-EphB mice for examining the importance of the tyrosine kinase activity of EphBs in both *in vitro* and *in vivo* settings. Our results demonstrate that PP1 analogs are capable of entering the brain tissue of an intact organism and then effectively and potently inhibiting EphB tyrosine kinase activity. These analogs also completely inhibited EphB tyrosine kinase activity in neuronal cultures, making it possible to use the AS-EphB TKI neurons to investigate the role of the kinase activity of EphBs in a diverse array of neuronal functions.

EphB RTK signaling mediates corpus callosum formation

Based on the finding that EphB tyrosine kinase activity mediates axon guidance at the optic chiasm, we next asked whether a similar mechanism might be used during other axon guidance decisions. We focused on the role of the tyrosine kinase activity of EphBs in the formation of the corpus callosum. Different EphB family members have been suggested to mediate this process

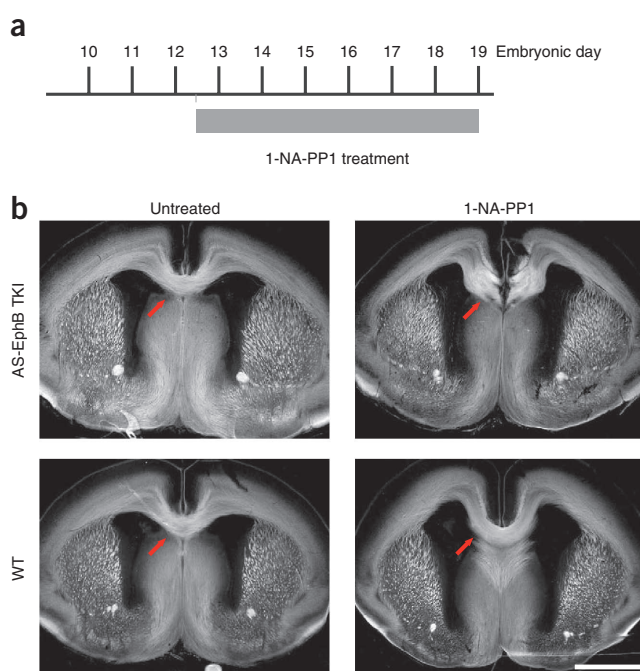


Figure 6 The kinase activity of EphBs is required for the formation of the corpus callosum *in vivo*. (a) Schedule of *in vivo* 1-NA-PP1 administration. Twice-daily subcutaneous injections of 80 mg per kg 1-NA-PP1 were administered to pregnant mice from E12.5–19. (b) Representative images of brain sections from E19 embryos stained with L1-CAM antibody (white) to visualize axon tracts. Red arrows denote the corpus callosum. Scale bar represents 1 mm.

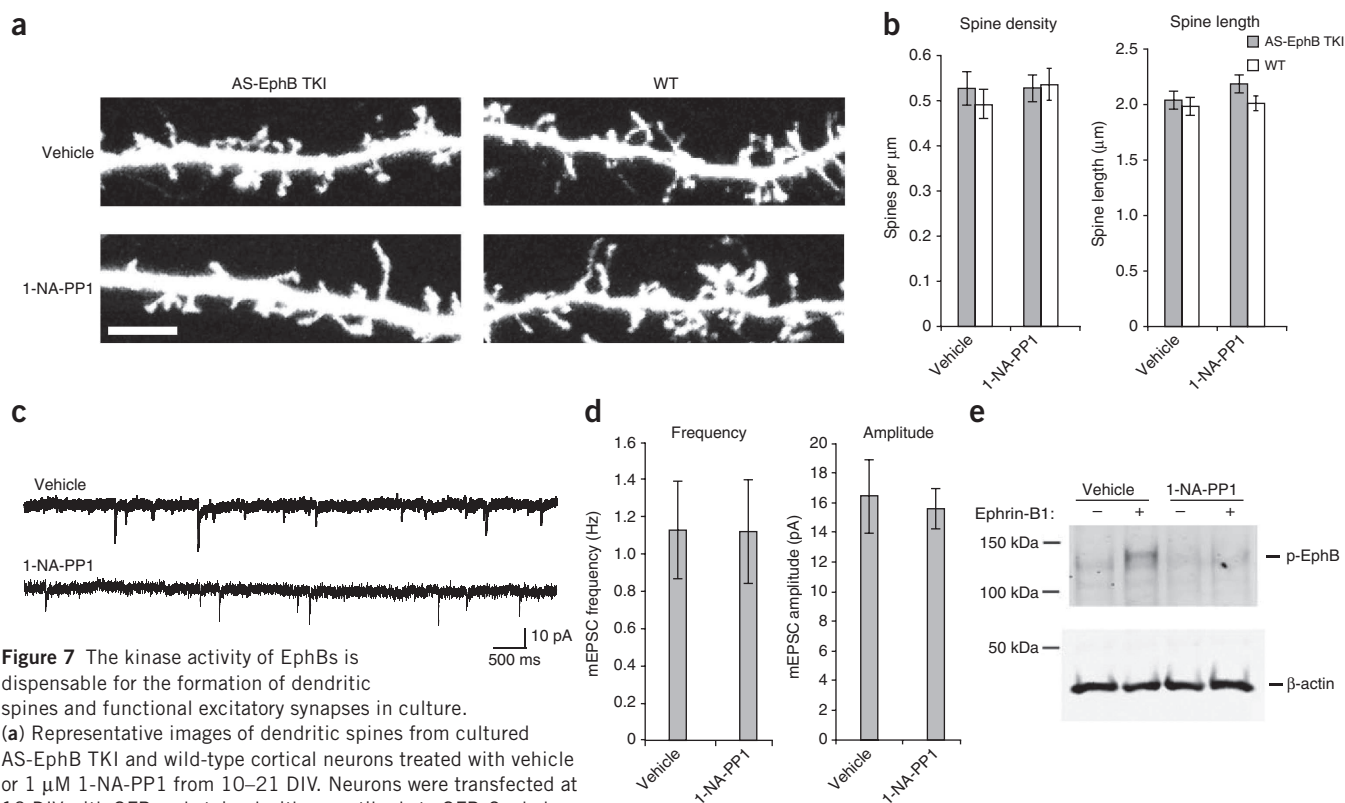


Figure 7 The kinase activity of EphBs is dispensable for the formation of dendritic spines and functional excitatory synapses in culture. **(a)** Representative images of dendritic spines from cultured AS-EphB TKI and wild-type cortical neurons treated with vehicle or 1 μM 1-NA-PP1 from 10–21 DIV. Neurons were transfected at 10 DIV with GFP and stained with an antibody to GFP. Scale bar represents 5 μm . **(b)** Quantification of spine density and spine length from **a**. Data are presented as mean $\pm$ s.e.m. $N = 25$ –33 neurons from independent biological replicates per condition. **(c)** Representative traces of recordings of mEPSCs from dissociated cortical neurons from AS-EphB TKI embryos at 10–12 DIV. Cultures were treated with vehicle or 1-NA-PP1 (1 μM) from 3 DIV until the time of recording. **(d)** Quantification of mEPSC frequency and amplitude from vehicle and 1-NA-PP1-treated neurons. Data are presented as mean $\pm$ s.e.m. $N = 13$ cells from independent biological replicates per condition. **(e)** The tyrosine kinase activity of AS-EphBs was inhibited by 1-NA-PP1. Cultures concurrent with those used in **d** were treated with 1-NA-PP1 under identical culture conditions then stimulated for 30 min with ephrin-B1 at 10 DIV. Cell lysates were analyzed by western blotting for phospho-EphB and β -actin. Uncropped blots are shown in **Supplementary Figure 7**.

through both forward and reverse signaling, but the requirement of the kinase activity has not been tested^{22,43}.

To address this question, we administered 1-NA-PP1 to pregnant AS-EphB TKI mice from E12.5 to E19.0 and visualized major axon tracts by LI-CAM staining (**Fig. 6a**). As expected, untreated AS-EphB TKI mice had normal corpus callosa (0 of 6 mice with agenesis; **Fig. 6b**). However, AS-EphB TKI mice treated with 1-NA-PP1 had partial corpus callosal agenesis with a gap in the dorsal midline region (11 of 11 mice with agenesis; **Fig. 6b**). Wild-type mice treated with 1-NA-PP1 had a normal corpus callosum (0 of 6 mice with agenesis), indicating that the corpus callosal agenesis phenotype was specific to inhibition of AS-EphBs (**Fig. 6b**). These data indicate that the tyrosine kinase activity of EphBs is essential for proper formation of the corpus callosum *in vivo*. Taken together with our findings on retinal axon guidance, these observations suggest that tyrosine kinase-dependent signaling is likely a general mechanism by which EphB forward signaling mediates repulsive axon guidance *in vivo*.

EphB RTK signaling is not required for synaptogenesis

In addition to their role in axon guidance, EphBs have been shown in numerous studies to control synapse development and function^{12,13,15,16,18,25,44,45}. Experiments involving overexpression of kinase-defective EphB mutants and the use of knock-in mice containing cytoplasmic truncations of EphBs have suggested a role for kinase signaling in synaptogenesis^{12,13,15,18,25}. However, the relevance of tyrosine

kinase activity of EphBs for synaptogenesis has not been examined under conditions in which ephrin-Bs and EphBs are expressed at endogenous levels and engage in physiological signaling.

The most notable synaptic defect observed after perturbation of EphB expression is the loss of dendritic spines, the sites on dendrites where most excitatory synapses form. In neurons from EphB1, EphB2 and EphB3 triple knockout mice, dendritic spine development is severely compromised^{13,18}. To assess the importance of EphB tyrosine kinase activity for spine development, we cultured cortical neurons from E15–17 AS-EphB TKI or wild-type embryos and treated them with 1-NA-PP1 (1 μM) between 10–21 DIV, the peak of spinogenesis. To ensure exposure to the full dose of the inhibitor during this long period, we changed the culture medium completely every 3–4 d with fresh 1-NA-PP1. To visualize the fine dendritic structures, we transfected neurons with GFP at 10 DIV and analyzed them at 21 DIV. To assess spine density, we counted spines per unit dendritic length over multiple segments of dendrite totaling more than 50 μm . Notably, we found that 1-NA-PP1 treatment had no detectable effect on spine number and length in AS-EphB TKI or wild-type neurons (**Fig. 7a,b**). This result suggests that the tyrosine kinase activity of EphBs is not required for dendritic spine development in dissociated cortical neurons.

To functionally test the effect of 1-NA-PP1 on synaptogenesis, we measured miniature excitatory postsynaptic currents (mEPSCs) in dissociated cortical neurons from AS-EphB TKI mice in the presence or absence of 1-NA-PP1. These cultures were treated with

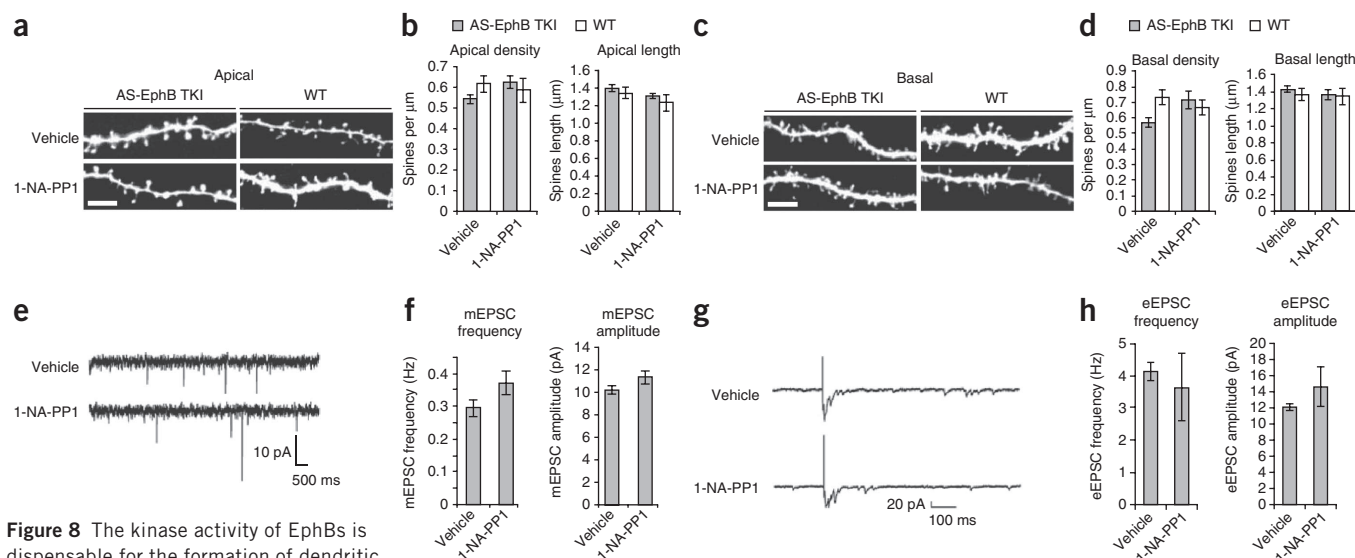


Figure 8 The kinase activity of EphBs is dispensable for the formation of dendritic spines and functional excitatory synapses in hippocampal slices. **(a)** Representative images of apical spines from AS-EphB TKI slices. Slices from postnatal day 5–7 (P5–7) mice were treated with vehicle or 1-NA-PP1 from 0–8 DIV. Scale bar represents 5 μm. **(b)** Quantification of apical spine density and spine length from **a** (wild-type + vehicle, $n = 21$ neurons; wild-type + 1-NA-PP1, $n = 11$ neurons; AS-EphB TKI + vehicle, $n = 30$ neurons; AS-EphB TKI + 1-NA-PP1, $n = 16$ neurons). **(c)** Representative images of basal spines from slices as described in **a**. **(d)** Quantification of basal spine density and spine length from **c**. **(e)** Representative traces from recordings of mEPSCs from P5–7 AS-EphB TKI slices at 12–14 DIV. **(f)** Quantification of mEPSC frequency and amplitude from **e**. (AS-EphB TKI + vehicle, $n = 16$ neurons; AS-EphB TKI + 1-NA-PP1, $n = 19$ neurons). 1-NA-PP1 did not produce a significant change in mEPSC frequency ($P = 0.11$) or amplitude ($P = 0.14$) by Student's t test. **(g)** Representative traces from recordings of eEPSCs from P5–7 AS-EphB TKI slices at 12–14 DIV. **(h)** Quantification of eEPSC frequency and amplitude in hippocampal slices. (AS-EphB TKI + vehicle, $n = 7$ neurons; AS-EphB TKI + 1-NA-PP1, $n = 7$ neurons). 1-NA-PP1 did not produce a significant change in eEPSC frequency ($P = 0.67$) or amplitude ($P = 0.33$) by Student's t test. All experiments were done in organotypic hippocampal slices. The concentration of 1-NA-PP1 was 1 μM. All data are presented as mean ± s.e.m. Samples are independent biological replicates.

1 μM 1-NA-PP1 from 3 to 10–12 DIV, and mEPSCs were measured by whole-cell electrophysiological recordings. Consistent with the spine analysis, we found no difference in mEPSC frequency or amplitude between AS-EphB TKI neurons treated with either vehicle or 1-NA-PP1 (**Fig. 7c,d**), which again strongly suggests that the tyrosine kinase activity of EphBs is not required for synaptogenesis, at least in dissociated neurons. To confirm that chronic 1-NA-PP1 treatment inhibits EphB tyrosine phosphorylation in these cultures, we performed western analysis on concurrent cultures and found that exposure of AS-EphB TKI neurons to 1-NA-PP1 (1 μM) resulted in a complete inhibition of ephrin-B-induced EphB autophosphorylation (**Fig. 7e**).

To determine whether the tyrosine kinase activity of EphBs might control excitatory synapse development or function in the context of a more intact neural circuit, we also examined the effect of 1-NA-PP1 inhibition on dendritic spine development, mEPSCs and evoked excitatory postsynaptic currents (eEPSCs) in organotypic hippocampal slices prepared from AS-EphB TKI mice. Expression of shRNAs targeting *Ephb1*, *Ephb2* and *Ephb3* led to pronounced alterations in spine development (**Supplementary Figs. 4a–c** and **5a–d**), and mEPSC and eEPSC currents (**Supplementary Fig. 6a–d**) in organotypic hippocampal slice cultures, indicating that EphBs are important for excitatory synaptic development under these conditions. However, in sharp contrast with these findings, the addition of 1-NA-PP1 to slices from AS-EphB TKI mice had no effect on dendritic spine growth (**Fig. 8a–d**), mEPSC characteristics (**Fig. 8e,f**) or eEPSC events (**Fig. 8g,h**). Taken together, these findings suggest that, at least under the conditions tested, EphB tyrosine kinase activity is not required for the formation or maintenance of functional synapses.

DISCUSSION

Employing a chemical genetic strategy, we engineered mice in which it is possible to acutely, reversibly and specifically inhibit the kinase signaling of EphBs *in vitro* and *in vivo*. We found that synaptogenesis, a process that requires EphB proteins, did not depend on the tyrosine kinase activity of EphBs. In contrast, we found a clear requirement for the tyrosine kinase activity of EphBs in ephrin-B-mediated growth cone collapse in culture and in the repulsive guidance of retinal and corpus callosal axons *in vivo*. Thus, unlike many other axon guidance receptors, EphBs mediate axon repulsion via a RTK-dependent mechanism.

Our finding that the kinase activity of EphBs is not required for the formation of excitatory synapses was unexpected and suggests a possible role for cytoplasmic domain oligomerization and other forms of protein-protein interactions in this process. For example, the binding of PDZ domain-containing synaptic proteins with the cytoplasmic tail of EphBs or the recruitment of NMDA receptors via the EphB extracellular region may initiate or stabilize the formation of excitatory synapses^{13,15}. Alternatively, EphBs could initiate signaling by recruiting other cytoplasmic kinases, such as the Src family of tyrosine kinases⁴⁴. Although our data strongly suggest that the tyrosine kinase activity of EphBs is not required for synaptogenesis under a wide range of experimental conditions tested, we cannot rule out the possibility that EphB kinase signaling has a role in other contexts. It will be important in the future to determine whether the kinase activity of EphBs functions instead in the plasticity of these synapses, as multiple reports have identified a role for EphBs in long-term potentiation in the hippocampus^{10,11}.

In contrast with synaptogenesis, our results reveal a requirement for the tyrosine kinase activity of EphBs in the formation of the ipsilateral retinal projection and the corpus callosum. Our finding

that EphB tyrosine kinase activity is required for retinal guidance is consistent with data from a previous study that found that overexpression of EphB1 is sufficient to drive retinal axons to ectopically project ipsilaterally and that this function requires an intact EphB1 kinase domain²³. These findings are also consistent with those of a recent study in which a knock-in mouse was generated with the intracellular region of EphB1 replaced with lacZ²⁴. This mouse displays a loss of the ipsilateral retinal projection, which indicates that EphB forward signaling is necessary for the formation of the ipsilateral retinal projection. Using AS-EphB TKI mice, we found that EphB tyrosine kinase activity is directly involved in the formation of the ipsilateral retinal projection.

Mutations of the cytoplasmic domains of EphBs or ephrin-Bs suggest that the formation of the corpus callosum is more complex than the ipsilateral retinal projection and could involve both EphB forward and reverse signaling²². However, our finding that inhibition of the kinase activity of EphBs *in vivo* results in a highly penetrant corpus callosum agenesis phenotype provides clear evidence for the requirement of EphB kinase signaling in the development of the corpus callosum.

On the basis of our observations, we favor a model in which EphB tyrosine kinase activity is required for repulsive interactions, such as in axon guidance, but may not be required for adhesive interactions, such as axon fasciculation and synapse formation. This model is consistent with previous theories that have suggested that the amount of kinase activation predicts the strength of repulsion⁴⁶. Furthermore, inhibiting kinase activity in a normally repulsive context (such as axon guidance) may lead to unnatural adhesion. Thus, it will be interesting to study the fate of the misprojected axons that we observed in the optic tract and corpus callosum. It will also be important to search for any counterexamples to our model, such as an adhesive interaction that is EphB tyrosine kinase dependent. The delicate balance between the opposing functionalities of the ephrin-B and EphB signaling system underscores the importance of studying these interactions under physiological conditions.

Understanding the downstream mechanism by which the tyrosine kinase activity of EphBs controls axon guidance represents an important direction for future studies. One crucial mediator of this signaling might be the Vav family of guanine nucleotide exchange factors for the small GTPase Rac. Vav2 is known to control growth cone collapse, and Vav2 and Vav3 double mutant mice display defects in the development of the ipsilateral retinocollicular projection^{39,47}. We found that the tyrosine kinase activity of EphBs was required for ephrin-B-induced Vav2 tyrosine phosphorylation. A thorough investigation of how EphBs control repulsive axon guidance will require knowledge of the full complement of tyrosine kinase substrates of EphBs. As analog-sensitive kinases accept orthologous ATP analogs that can directly label the targets of the kinase⁴⁸, it should be possible to use AS-EphB TKI mice to identify direct kinase substrates of EphBs. Given the general limited knowledge of axon guidance mechanisms, identification of EphB substrates in the relevant neurons could be a powerful approach for uncovering these mechanisms.

Our chemical genetic approach offers several advantages over conventional genetic loss-of-function studies⁴⁹. Given that we were able to block the kinase activity of EphBs while leaving their scaffolding and reverse signaling capabilities intact, it was possible to dissect the role of the kinase activity of EphBs *in vivo* under conditions in which EphBs were expressed at physiological levels. As there are no Cre/*loxP*-based conditional EphB mice of any kind available, AS-EphB TKI mice represent an alternative for many experiments that require conditional regulation of EphB signaling. In addition, the reversible nature and the fine temporal control afforded by the chemical genetic

approach should permit investigations into the functions of EphBs in the mature animal, such as in adult neurogenesis and synaptic plasticity, and in pathologies such as Alzheimer's disease, autism and cancer^{4,7,50}. This new window into EphB signaling should also provide insights that are crucial for therapeutic drug development for the treatment of EphB-mediated disease.

METHODS

Methods and any associated references are available in the [online version of the paper](#).

Note: Supplementary information is available in the online version of the paper.

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AUTHOR CONTRIBUTIONS

M.J.S., H.-Y.H.H. and M.E.G. conceived and designed the study. M.J.S. and H.-Y.H.H. conducted all of the experiments unless otherwise noted. B.L.B. and N.S. performed electrophysiological recordings. J.Z. generated EphB1 and EphB3 targeting constructs. M.A.R. and C.W.C. contributed to the axon guidance experiments. A.N.M. and A.A.R. generated shRNAs for EphBs. B.A. performed qPCR experiments. C.Z. and K.M.S. designed and synthesized inhibitors. M.J.S., H.-Y.H.H. and M.E.G. wrote the manuscript.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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ONLINE METHODS

Animals. *EphB1*^{T697G}, *EphB2*^{T699A} and *EphB3*^{T706A} single mutants were generated individually by homologous recombination in mouse ES cells. Mice harboring each of the knock-in mutations were intercrossed to obtain the triple homozygous AS-EphB TKI mice.

To generate the targeting constructs, the 5' and 3' arms was PCR amplified from J1 ES cell DNA using primers listed in **Supplementary Table 1** and subcloned into a modified pKSNeoDTA vector (originally constructed in the laboratory of P. Soriano, Mount Sinai Hospital) containing a *loxP-neo-loxP* cassette for positive selection and a diphtheria toxin A negative-selection cassette (DTA). The analog-sensitive mutations were introduced by site-directed mutagenesis. The targeting constructs were confirmed by sequencing. Linearized targeting construct was electroporated into J1 ES cells, which were subsequently selected with G418. Correct targeting of ES cells was initially screened by PCR and then confirmed by Southern analysis and direct sequencing of PCR products amplified from the mutated alleles. The positive clones were karyotyped and the neomycin resistance cassette was removed by electroporating targeted ES cells with a Cre expression plasmid. ES cell clones were micro-injected into C57BL/6 blastocysts to generate chimeric mice. Male chimeric animals were mated to C57BL/6 wild-type females for germline transmission of the targeted allele.

The ES cell targeting efficiencies were as follows: EphB1 knock-in (17 of 192, 9%), EphB2 knock-in (3 of 186, 2%) and EphB3 knock-in (12 of 96, 13%).

Mice were maintained as homozygotes in a mixed 129 and C57BL/6 background. Unless noted, wild-type mice were F1 offspring of a C57BL/6 × 129sv cross. Animals were housed under a 12-h light/dark cycle. No more than five animals were housed in each cage. Mice and embryos were chosen at random, regardless of sex, for treatment condition.

All experiments with mice were approved by the Animal Care and Use Committee of Harvard Medical School. E0 was defined as midnight preceding the morning a vaginal plug was found.

HEK 293 cell culture and transfection. HEK 293 cells were maintained in DMEM supplemented with 10% fetal bovine serum (vol/vol, Gibco), 2 mM glutamine (Gibco) and penicillin/streptomycin (100 U ml⁻¹ and 100 µg ml⁻¹, respectively; Gibco). HEK 293 cells were transfected using the calcium phosphate method, as previously described¹⁵.

Antibodies and western blotting. The following antibodies were purchased commercially: pan-phospho-tyrosine pY99 mouse monoclonal (Santa Cruz), EphB1 (H-80) and EphB3 (H-85) rabbit polyclonal (Santa Cruz), β-actin mouse monoclonal (Abcam), PSD-95 mouse monoclonal (Pierce), and synapsin rabbit polyclonal (Millipore). Mouse monoclonal antibody to neurofilament (2H3) was obtained from the Developmental Studies Hybridoma Bank (T. Jessell, Columbia University). Polyclonal rabbit antibodies to Vav2, pan-phospho-EphB and EphB2 (all at 1:1,000 dilution) were generated in M.E.G.'s laboratory and were described previously¹⁵. The phospho-specificity of the pan-phospho-EphB antibody was validated in HEK 293 cells for EphB1, EphB2 and EphB3 individually. All western blots were imaged and quantified using the Odyssey Infrared Imaging System (Licor) using fluorescently labeled secondary antibodies (Rockland Immunochemicals).

Protein binding assays. Full-length human Grb2 and rat Pick1 were PCR amplified from plasmids #26085 and #31613 (Addgene), respectively. Grb2 and Pick1 were fused to GST at their N termini by cloning them into a pGEX vector (Pharmacia), expressed in *E. coli*, and affinity purified on glutathione sepharose beads. GST, used as a negative control, was expressed from the empty pGEX vector. AS-EphB1 protein was expressed in HEK 293 cells by transient transfection and the cells were treated with either DMSO or 1 µM 1-NA-PP1 for 16 h. Cells were lysed in lysis buffer (30 mM HEPES (pH 7.7), 100 mM KCl, 1 mM MgCl₂, 2 mM DTT, 2 mM sodium orthovanadate, 1% Triton X-100 (vol/vol), protease inhibitor (Roche, 04693159001) and phosphatase inhibitor cocktails (Sigma, P5726, P0044)). The crude lysates were centrifuged at 70,000g for 20 min at 4 °C to generate the high-speed supernatants. High-speed supernatant was incubated with glutathione beads coated with approximately GST, GST-Grb2 or GST-Pick1 with rotation at 4 °C for 2 h. The beads were washed three times with lysis buffer and analyzed by SDS-PAGE and western blotting.

Pharmacology. 1-NA-PP1 was synthesized as described previously²⁷ and dissolved in DMSO. 3-MB-PP1 was synthesized using a similar procedure and dissolved in DMSO. Dose-response curves using 1-NA-PP1 and 3-MB-PP1 were calculated on GraphPad Prism using the least-squares method. The vehicle dose was calculated as two orders of magnitude below the lowest dose (0.05 nM). For wild-type EphBs, 100% inhibition was defined at 1 mM.

Neuronal cell culture. Cortical and hippocampal neurons were prepared from E15–17 mouse embryos as previously described¹⁵. Cultured neurons were maintained in Neurobasal medium (Invitrogen) supplemented with 1× B27 (Invitrogen), penicillin/streptomycin (100 U ml⁻¹ and 100 µg ml⁻¹, respectively) and 2 mM glutamine. For biochemistry, neurons were seeded at a density of 2 × 10⁶ neurons per well of a six-well plate coated with polyornithine (Sigma). For electrophysiology and imaging, neurons were seeded at a density of 7.5 × 10⁴ neurons per well on a glial monolayer on glass coverslips coated with polyornithine and laminin (Invitrogen).

Organotypic slices. Hippocampal organotypic slices were prepared in ice-cold dissection media (1 mM CaCl₂, 5 mM MgCl₂, 10 mM D-glucose, 4 mM KCl, 26 mM NaHCO₃, 218 mM sucrose, 1.3 mM sodium phosphate and 30 mM HEPES, pH 7.4). Brains were isolated from P5–7 pups, and hippocampi were excised and chopped into 400-µm sections. Slices were cultured on Millicell cell culture inserts (Millipore) in media containing 20% horse serum, 1 mM L-glutamine, 0.0012% ascorbic acid (wt/vol), 1 µg ml⁻¹ insulin, 1 mM CaCl₂, 2 mM MgCl₂, 2.3 mg ml⁻¹ glucose, 0.44 mg ml⁻¹ NaHCO₃ and 7.16 mg ml⁻¹ HEPES in MEM.

Ephrin stimulation. For ephrin stimulations in dissociated cultured neurons and retinal explants, mouse ephrin-B1-Fc or ephrin-B2-Fc (R&D Systems) was pre-clustered for 50 min with goat antibody to human IgG Fc (Jackson ImmunoResearch, 109-001-008) at 22–25 °C in phosphate-buffered saline (PBS) at a molar ratio of 1:1 before stimulation. Pre-clustered ephrin-B1-Fc or ephrin-B2-Fc was added to the appropriate medium at a final concentration of 2.5 µg ml⁻¹. As a control, clustered human Fc in media was applied to neurons where specified.

Cell lysis and immunoprecipitation. Cultured cells were collected and homogenized in RIPA buffer (50 mM Tris (pH 8.0), 150 mM NaCl, 1% Triton X-100, 0.5% sodium deoxycholate, 0.1% SDS (wt/vol), 10 mM NaF, complete protease inhibitor cocktail tablet (Roche), 1 mM sodium orthovanadate, and phosphatase inhibitor cocktails 1 and 2 (1×, Sigma)). After clearing lysates, supernatants were incubated with the appropriate antibody for 1 h at 4 °C, followed by addition of Protein-A Fastflow agarose beads (Sigma) for 1 h. Beads were washed in lysis buffer or PBS three times and eluted in 2× SDS sample buffer followed by boiling.

Surface labeling. Labeling of surface proteins was performed using the Pierce Cell Surface Protein Isolation Kit (Thermo Scientific). After chronic treatment with vehicle or 1 µM 1-NA-PP1, cultured cortical neurons were incubated with EZ-link biotin for 30 min at 22–25 °C, washed with PBS, and lysed in RIPA buffer. Lysates were immunoprecipitated with an antibody to EphB2 or a control antibody, and probed with either antibody to EphB2 or fluorescently labeled streptavidin (Invitrogen).

Retinal explants. Ventrotemporal segments of retina were microdissected from E14.5 mouse embryos and cultured as previously described⁵¹. Embryos were removed from the uterus and decapitated, and heads were placed in ice-cold DMEM/F12 (Gibco). Ventrotemporal sections of the retina were excised and placed on glass coverslips coated with polyornithine and laminin. Explants were maintained in serum free medium (10 mg ml⁻¹ BSA (Sigma), 1% ITS supplement (vol/vol) (Sigma), penicillin/streptomycin (20 U ml⁻¹ and 20 µg ml⁻¹, respectively, Gibco) in DMEM/F12) supplemented with 0.2% methyl cellulose (wt/vol, Sigma) to increase media viscosity and minimize explant movement. All experiments were conducted 18–24 h after initial plating.

Immunocytochemistry and growth cone collapse assay. For experiments with inhibitor treatment, explants were pre-incubated with vehicle or PP1 analogs for 1 h, followed by a 30-min ephrin-B2 stimulation, with variations as described in the text. Following stimulation, retinal explants were fixed for 20 min at 25 °C with 4% paraformaldehyde (PFA)/2% sucrose (wt/vol) in PBS. Explants were

then blocked in 10% goat serum (vol/vol), 0.2% Tween-20 (vol/vol) in PBS for 1 h, followed by incubation with antibody to either neurofilament phospho-EphB in 50% blocking solution overnight. After PBS washes, explants were incubated in Alexa Fluor-conjugated secondary antibodies (Invitrogen) and Alexa Fluor 488-conjugated phalloidin (Invitrogen). Explants on coverslips were mounted on glass slides using Fluoromount-G (Southern Biotech). Neurons were imaged using a laser-scanning Zeiss Pascal microscope using a 40× objective with sequential acquisition settings at 1,024 × 1,024 pixel resolution. All imaging and image analysis were performed blind to the genotype and treatment condition of the samples. At least ten growth cones were analyzed per explant.

In vivo 1-NA-PP1 delivery. Pregnant wild-type or AS-EphB TKI mice were injected subcutaneously twice daily with 80 mg per kg of body weight 1-NA-PP1 dissolved in 10% DMSO, 20% Cremaphor-EL and 70% saline (vol/vol) from E13.5–16.5 for optic tract experiments or from E12.5–19 for cortical tract experiments. All experiments included data from at least two separate litters of embryos per condition. Animals used for *in vivo* 1-NA-PP1 treatment had no prior exposure to 1-NA-PP1 or other drugs.

DiI labeling. DiI labeling was performed as previously described⁵². At E16.5, embryo heads were fixed in 4%PFA/2% sucrose in PBS overnight and then washed with PBS. The lens and retina were removed from the left eye and a small crystal of DiI (Invitrogen) was placed in the optic disc. The retina was then replaced securely and the heads were stored in PBS + 0.1% azide (wt/vol) at 22–25 °C for 12 d. After labeling, brains were removed and fluorescent optic tracts were imaged on a Leica MZ16F fluorescent stereomicroscope. Images were captured using Spot Advanced software. Labeling was quantified using Metamorph software by drawing rectangular regions of interest around the ipsilateral and contralateral tracts, subtracting background, and calculating the ipsilateral index on the basis of the integrated intensity of fluorescence: ipsilateral index = ipsilateral / (ipsilateral + contralateral). To compare wild-type and AS-EphB TKI responses with respect to 1-NA-PP1 treatment, each genotype was normalized to its untreated condition, producing a normalized ipsilateral index.

Analysis of corpus callosum phenotypes. E19.0 embryos were fixed in 4% PFA/2% sucrose in PBS for 2 d, then stored in PBS + 0.02% sodium azide at 4 °C. Brains were removed and vibratome sectioned to 70 μm. Sections were blocked in 5% normal donkey serum, 1% BSA, 0.2% glycine (wt/vol), 0.2% lysine (wt/vol) with 0.3% Triton X-100 in PBS at 22–25 °C for 1 h. To stain axon tracts, sections were incubated with rat antibody to L1-CAM (Millipore, MAB5272MI) at a 1:200 dilution in blocking solution at 4 °C overnight. Sections were washed three times with PBS and incubated with Cy3-conjugated donkey antibody to rat IgG (Jackson) for 1.5 h. Sections were washed 7–8 times with PBS and mounted with Aqua-mount (Lerner laboratories). Sections were imaged on an Olympus Bx51 epifluorescent microscope at 4× magnification. Corpus callosum partial agenesis was scored as the apparent failure of corpus callosal axons to cross the midline in a specific region along the rostral-caudal brain axis, and was evident by a gap in the dorsal midline region.

qPCR. Total RNA was isolated from mouse 7 DIV cortical cultures using Trizol reagent (Invitrogen) according to the manufacturer's instructions. Isolated RNA was treated with DNaseI Amplification Grade (Invitrogen) and a cDNA library was synthesized by cDNA High Capacity cDNA Reverse Transcription Kit (Applied Biosystems). The cDNA was the source of input for qPCR, using a Step One Plus Real-Time PCR Instrument and SYBR Green reagents (Applied Biosystems). The relative expression plot was generated using concentration values that were normalized to corresponding actin concentrations. The following qPCR primer pairs were used: EphB1 forward, ACTGCAGAGTTGGGATGGAC; EphB1 reverse, CATCATAGCCACTGACTTCTTCC; EphB2 forward, TTCATGGAGAACGGATCTCTG; EphB2 reverse, GACTGTGAAGTCCCCATCG; EphB3 forward, CCCTGGACTCCTTCTTACGG; EphB3 reverse, GCAATGCC TCGTAACATGC.

Analysis of dendritic spines. For dissociated neuron experiments, cortical neurons were cultured as described above and grown on a monolayer of astrocytes on glass coverslips. Neurons were treated with vehicle or PP1 analogs at 10 DIV

and media was changed entirely every 3–4 d (using neuronally pre-conditioned media). Neurons were transfected with GFP at 10 DIV using Lipofectamine 2000 and fixed with 4% PFA/2% sucrose in PBS at 21 DIV. Cells on coverslips were stained for GFP and mounted on slides using Fluoromount-G.

Neurons were imaged on a Zeiss Pascal confocal microscope, using a 63× objective, and maximal z projections were analyzed using Metamorph software. Multiple sections of dendrite totaling >50 μm were counted for each neuron.

For experiments in slice, hippocampal slices were treated with vehicle or PP1 derivatives at 2 DIV and media/drug were fully replaced every 2–3 d. At 2–3 DIV, plasmids encoding GFP (or GFP and EphB shRNAs) was biolistically transfected using a Helios gene gun. DNA bullets were prepared from 1.6-μm gold micro-carrier particles (Biorad). After 8–9 DIV, slices were fixed in 3.2% PFA/5% sucrose in PBS for 1 h and stained with chicken antibody to GFP (Aves Labs) and rabbit antibody to NeuN (Millipore, clone A60, 1:1,000 dilution) to visualize the structure of hippocampal fields. Basal and apical dendrites were analyzed separately and sections of dendrite totaling >50 μm were counted for each neuron.

Electrophysiology. For experiments in dissociated neurons, whole-cell voltage-clamp recordings were obtained using an Axopatch 200B amplifier at 25 °C. During recordings, neurons were perfused with artificial cerebrospinal fluid containing 127 mM NaCl, 25 mM NaHCO₃, 1.25 mM Na₂HPO₄, 2.5 mM KCl, 2 mM CaCl₂, 1 mM MgCl₂, 25 mM glucose, and saturated with 95% O₂, 5% CO₂. Vehicle or 1-NA-PP1 treatment was initiated at 3 DIV and continued throughout recordings. The internal solution used in all electrophysiological experiments contained 120 mM cesium methane sulfonate, 10 mM HEPES, 4 mM MgCl₂, 4 mM Na₂ATP, 0.4 mM Na₂GTP, 10 mM sodium phosphocreatine and 1 mM EGTA. Osmolarity and pH were adjusted to 310 mOsm and 7.3 with Millipore water and CsOH, respectively.

mEPSCs were isolated by exposing neurons to 0.5 μM tetrodotoxin, 50 μM picrotoxin, and 10 μM cyclothiazide (all from Tocris Bioscience). Cells with series resistance larger than 25 MΩ during the recordings were discarded. Data were analyzed in IgorPro (Wavemetrics) using custom-written macros. For each trace, the event threshold was set 1.5-fold greater than the root-mean-square current. Currents were counted as events if they crossed the event threshold, had a rapid rise time (1.5 pA ms⁻¹) and had an exponential decay ($\tau < 50$ ms for mEPSC).

As a control for inhibition of EphBs, concurrent plates of neurons were treated with inhibitor and at the time of recording were stimulated with ephrin-B1 for 30 min. Neurons were lysed in 1× SDS-sample buffer, run on western blot and probed with rabbit antibody to phospho-Eph and mouse antibody to β-actin (Abcam).

For mEPSC experiments in organotypic hippocampal slices, whole-cell voltage-clamp recordings were made from visually identified CA1 pyramidal neurons and the mEPSC amplitude and frequency measured. Slices were treated with vehicle or 1-NA-PP1 from 2 DIV until the time of recording.

To evaluate evoked synaptic transmission, the Schaffer collaterals were depolarized with an extracellular stimulating electrode and the postsynaptic eEPSC response measured from CA1 neurons⁵³. In these experiments, the artificial cerebrospinal fluid contained 4 mM Sr²⁺ instead of CaCl₂ and 4 mM MgCl₂ so that the extracellular stimulation resulted in asynchronous presynaptic vesicle fusion. The stimulus strength was set so that the initial postsynaptic response was 50–100 pA and the current amplitude and frequency of the asynchronous EPSCs occurring 400–900 ms post-stimulation was measured. Slices were treated with vehicle or 1-NA-PP1 from 2 DIV until the time of recording. Analysis of eEPSCs was performed using custom-written macros in IgorPro.

Statistical analysis. All animal experiments contained pups from multiple litters. All imaging analyses were done blind to condition. No data points were excluded in any experiment. Unpaired *t* tests and two-way ANOVA (for comparing effect of drug on AS EphB TKI versus wild-type neurons) analyses were conducted using GraphPad Prism software. All tests are two-sided (standard).

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